

Title :

AI-Powered Learning Management System (AI-LMS)

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Roll No : 03

KTE25MCA - 2003

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PROJECT WORK LOG & PROGRESS RECORD

• Date : 11 - 07 - 2026

- Installed the required software tools (Node.js, Express.js, MongoDB, React, Vite).
- Initialized React.js framework for frontend web application development.
- Organized all project development tools, dependencies, and environment configurations.

• Date : 15 - 07 - 2026

- Planned implementation of the project architecture and technology stack.
- Selected core backend and frontend modules for the AI-LMS project.
- Setup developer workflow, version control, and coding standards.

• Date : 19 - 07 - 2026

- Noted down detailed database requirements and entity attributes.
- Discussed entity relationships between database tables/collections (Users, Courses, Enrollments).
- Started designing the MongoDB Mongoose database structure.

• Date : 21 - 07 - 2026

- Planned the project folder structure (client/ and server/ separation).
- Created folders for frontend, backend, models, controllers, services, and documentation.
- Created README.md file and updated initial project details.
- Initialized Git repository and pushed project baseline files to GitHub.

• Date : 22 - 07 - 2026

- Finalized the project scope and identified major project modules:
 1. Authentication & Authorization
 2. User Management
 3. Course Management
 4. Learning Content Management
 5. Assignment & Exam Management
 6. AI Tutor (RAG)
 7. Personalized Learning
 8. Study Planner
 9. Performance Prediction
 10. Analytics Dashboard & Reports
- Planned the project workflow and role permissions (Admin, Faculty, Student).
- Started designing the complete Database ER diagram.

• Date : 25 - 07 - 2026

- Completed Module 1: Authentication & Authorization.
- Implemented Student Registration, Login, JWT Authentication, and bcrypt password hashing.
- Implemented Forgot Password and Reset Password security token workflows.
- Created React ProtectedRoute and RoleGuard components for client-side route protection.

• Date : 27 - 07 - 2026

- Completed Module 2: User Management.
- Developed Administrative User Management interface (Manage Students, Manage Faculty, View, Search, Filter).
- Implemented User Activation / Deactivation status toggles and account deletion capabilities.
- Created centralized database seeding script to seed default Admin, Faculty, and Student accounts.

• Date : 29 - 07 - 2026

- Planned and designed Module 3: Course Management.
- Designed normalized Mongoose schemas for Course, Category, and Enrollment collections.
- Enforced strict public registration rules (Student self-registration only; Admin-controlled Faculty creation).
- Built clean Notion/Linear-inspired modern UI for Course Catalog, Course Detail, and Manage Courses views.

- **Date : 02 - 08 - 2026**

- Completed Module 4: Learning Content Management.
- Built Course Section and Lesson management APIs supporting CRUD and reordering.
- Implemented file upload capability for course documents and slides (PDF, docx, videos).
- Developed interactive Lesson Viewer interface on the client with embedded PDF renderer.

- **Date : 06 - 08 - 2026**

- Completed Module 5: Assignment & Exam Management.
- Developed assignment upload, student submission portal, and grading queue for faculty.
- Designed online Quiz Engine supporting MCQ and True/False questions with student score persistence.
- Built interactive quiz taking UI with a live count-down timer and automatic scoring.

- **Date : 10 - 08 - 2026**

- Completed Module 6: AI Tutor (RAG).
- Configured text extraction from lecture materials and background indexing pipeline.
- Integrated @google/genai SDK for native, robust Gemini AI chat completions.
- Built similarity search filters on Pinecone database and interactive UI with citation badges.

- **Date : 14 - 08 - 2026**

- Completed Module 7: Personalized Learning.
- Implemented student performance analyzer telemetry based on quiz and assignment attempts.
- Developed automatic weak-topic detection and custom recommendation engine.
- Created responsive UI listing personalized concept reviews, weak topics, and study badges.

- **Date : 17 - 08 - 2026**

- Completed Module 8: Study Planner.
- Developed study plan schema and backend scheduler service.
- Integrated AI study timetable generator with fallback algorithmic scheduler.
- Designed interactive calendar view displaying tasks, exam countdowns, and completion statuses.

- **Date : 20 - 08 - 2026**

- Completed Module 9 & 10: Performance Prediction, Analytics, and Final Integration.
- Aggregated platform statistics and developed consolidated dashboards for Admin, Faculty, and Students.
- Resolved database schema validation issues (fixed Mongoose CastError on generatedQuestions).
- Conducted full system verification and end-to-end user acceptance testing workflows.